

Multi-Purpose Wire-Loop SECURITY SYSTEM

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This multi-purpose security system utilises wire loops to secure items. Wire loops can be employed in various ways for security applications such as luggage security, bike security, bicycle security, room security, and door security. A wire loop, when attached to security items and alarm mechanisms, detects and alerts in case of unauthorised attempts to tamper with the wire loop.

Use a thin, flexible, and durable conductive wire to create a loop around the area you want to secure. Install the wire loop around the desired perimeter, whether it be a doorway, window, or any other access point. When a breach is detected (e.g., someone cuts or breaks the wire loop), the alarm system activates.

Circuit and working

The circuit diagram of the multi-purpose security system is shown in Fig. 2. It is built around transistor BC327 (T1), melody generator UM66 (IC1), low-power audio amplifier LM386 (IC2), 8-ohm half-watt loudspeaker (LS1), 9V battery or 9V adaptor, and a few other components. Around one metre of flexible wire is used as a sensor, forming a loop to connect the article (such as luggage) to CON1 in the circuit.

The melody signal produced by the UM66 melody generator is weak. To amplify the sound signals, the widely-used audio amplifier LM386 is employed.

To understand the working of

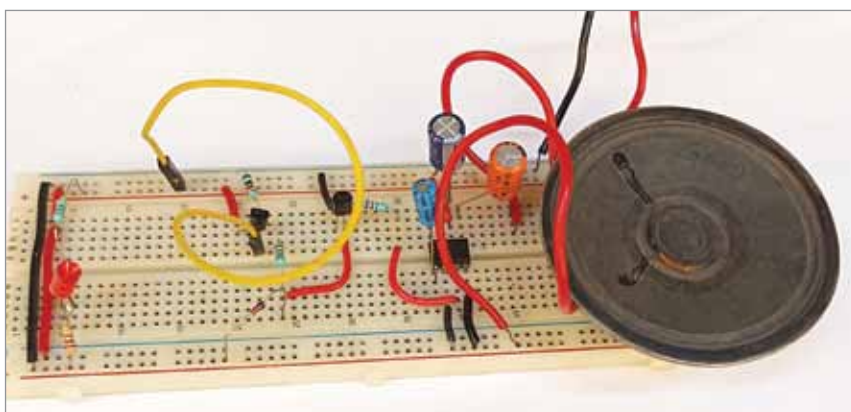


Fig. 1: Author's prototype on breadboard

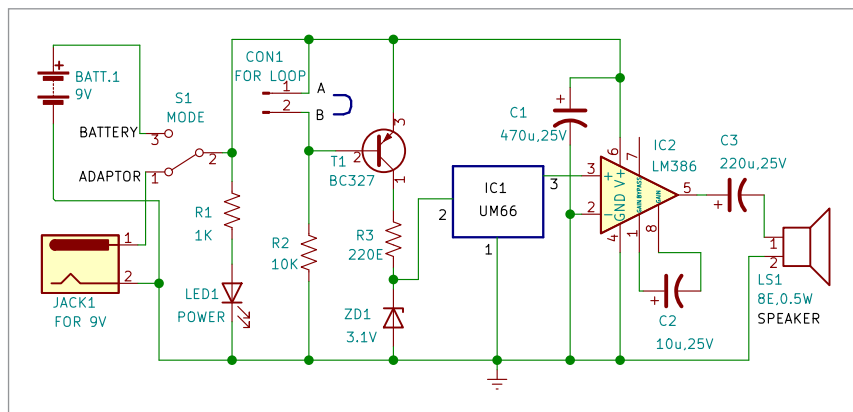


Fig. 2: Circuit diagram

the circuit, first comprehend the pin descriptions of LM386. Pins 1 and 8 are used for gaining control of the amplifier. Pins 2 and 3 feed the input signal for amplification, with pin 2 for negative input and pin 3 for positive signal. Pin 4 is used for ground, while pin 6 is used for the positive 9V DC power supply. Pin 5 is the output, connected to a loudspeaker through a capacitor. Pin 7 is not utilised in this configuration.

The system's operation is straightforward. When power is supplied via a battery or adaptor, the musical IC UM66 (IC1) gets activated and begins producing musical sound at its output pin 3. The speaker can be directly attached between pin 3 and ground of IC1 to hear the musical sound. For louder sound, LM386 is used. The 3.1V zener diode ZD1 is connected at pin 2 of IC1 to protect it from voltage beyond 3.1V.